Assignment 1 Grading Rubric

IEDA 1250: Optimization for Decision Making (Summer, 2026)

Question 1: Linear Programming – Graphical Solutions (10 Points)

- Correctly analyzes LP (a), including the graphical feasible region or equivalent reasoning, and concludes that the LP is unbounded. **(2.5 Points)**
- Correctly analyzes LP (b), including the graphical feasible region or equivalent reasoning, and concludes that the LP is unbounded. **(2.5 Points)**
- Correctly analyzes LP (c), identifies a unique optimal solution

  $$x_1^* = 1, \qquad x_2^* = 0,$$

  and reports the corresponding objective value. **(2.5 Points)**
- Correctly analyzes LP (d), identifies infinitely many optimal solutions on

  $$x_1 + 3x_2 = 3$$

  between

  $$\left(\frac{3}{2}, \frac{1}{2}\right) \quad \text{and} \quad (3, 0),$$

  and reports the corresponding objective value. **(2.5 Points)**

Question 2: Duality – Transforming a General LP to Standard Form (10 Points)

- Correctly substitutes

  $$x_1 = -\bar{x}_1, \qquad x_2 = \bar{x}_2 - \underline{x}_2,$$

  with all new variables nonnegative, and rewrites the objective and constraints in the new variables. **(2 Points)**
- Correctly replaces the equality constraint with one "$\leq$" constraint and one "$\geq$" constraint. **(2 Points)**
- Correctly converts all "$\geq$" inequalities to "$\leq$" inequalities so that the primal is in a form with only "$\leq$" constraints and nonnegative variables. **(2 Points)**
- Correctly writes the dual of the transformed LP, including the correct objective, constraints, and nonnegativity restrictions on the dual variables. **(2 Points)**
- Correctly simplifies the dual LP to the equivalent compact form with one free dual variable and one nonnegative dual variable. **(2 Points)**

Question 3: Complementary Slackness (10 Points)

- Correctly writes or uses the dual LP:

$$\max \quad 4y_1 + 3y_2$$

  with the five dual constraints corresponding to $x_1, \ldots, x_5$ and $y_1, y_2 \geq 0$. **(2 Points)**

- Correctly evaluates the dual constraints at

$$y_1^* = 0.8, \qquad y_2^* = 0.6,$$

  and identifies which dual constraints are tight and which have slack. **(2 Points)**

- Correctly applies complementary slackness to conclude

$$x_2^* = x_3^* = x_4^* = 0.$$

  (2 Points)

- Correctly uses $y_1^* > 0$ and $y_2^* > 0$ to make both primal constraints tight and solves

$$x_1 + 3x_5 = 4, \qquad 2x_1 + x_5 = 3.$$

  (2 Points)

- Correctly reports the optimal primal solution and objective value:

$$x_1^* = 1, \quad x_2^* = 0, \quad x_3^* = 0, \quad x_4^* = 0, \quad x_5^* = 1, \qquad z^* = 5.$$

  (2 Points)

Question 4: Linear Programming Model and Excel Solution – PaGas Water Operation (10 Points)

- Correctly defines and uses the five decision variables $X_{AF}, X_{BF}, X_{AR}, X_{BR}, X_F$. **(1 Points)**

- Correctly formulates the total cost objective:

$$\min \quad 200X_{AF} + 100X_{AR} + 180X_{BF} + 120X_{BR} + 70X_F.$$

  (2 Points)

- Correctly includes the water balance constraints at Plant A, Plant B, and the fracking site:

$$X_{AF} + X_{AR} = 20, \quad X_{BF} + X_{BR} = 25, \quad X_{AF} + X_{BF} + X_F = 50.$$

  (3 Points)

- Correctly imposes nonnegativity on all decision variables. **(1 Points)**

- Provides a valid Excel Solver setup or screenshot showing the model and optimal solution. **(1 Points)**

- Correctly reports the optimal solution:

$$X_{AF}^* = 0, \quad X_{AR}^* = 20, \quad X_{BF}^* = 25, \quad X_{BR}^* = 0, \quad X_F^* = 25.$$

  (2 Points)

Question 5: Binary Integer Programming – Crew Assignment (20 Points)

- Correctly defines twelve binary variables

$$x_i = \begin{cases} 1, & \text{if sequence } i \text{ is selected,} \\ 0, & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, 12.$$

  (2 Points)

- Correctly formulates the cost-minimization objective:

$$\min \ 2x_1 + 3x_2 + 4x_3 + 6x_4 + 7x_5 + 5x_6 + 7x_7 + 8x_8 + 9x_9 + 9x_{10} + 8x_{11} + 9x_{12}.$$

  (3 Points)

- Correctly writes coverage constraints requiring each flight to be covered by at least one selected sequence. Award partial credit for each correctly modeled flight coverage constraint. **(7 Points)**

- Correctly imposes the exactly-three-crews constraint:

$$\sum_{i=1}^{12} x_i = 3.$$

  (2 Points)

- Correctly states the binary restrictions $x_i \in \{0, 1\}$ for all $i = 1, \ldots, 12$. **(2 Points)**

- Provides a valid Excel Solver setup or screenshot showing the integer programming model and optimal solution. **(2 Points)**

- Correctly reports one optimal solution and its total cost. Valid optimal solutions include

$$x_3 = x_4 = x_{11} = 1$$

  or

$$x_1 = x_5 = x_{12} = 1,$$

  with all other variables equal to 0 and total cost 18,000. **(2 Points)**

Question 6: Mixed Integer Reformulation of Logical Constraints (10 Points)

- Correctly introduces binary variables to indicate whether each logical inequality is enforced. **(2 Points)**

- Correctly reformulates part (a), "at least one of two inequalities holds," using a valid big-M formulation, for example:

$$3x_1 + x_2 + x_3 + x_4 \le 12 + M(1 - y_1),$$

$$x_1 + x_2 + x_3 + x_4 \le 15 + M(1 - y_2),$$

$$y_1 + y_2 \ge 1, \qquad y_1, y_2 \in \{0, 1\}.$$

  (4 Points)

- Correctly reformulates part (b), "at least two of three inequalities hold," using a valid big-M formulation, for example:

$$2x_1 + 5x_2 + x_3 + x_4 \leq 30 + M(1 - y_1),$$
$$x_1 + 3x_2 + 5x_3 + x_4 \leq 40 + M(1 - y_2),$$
$$3x_1 + x_2 + 3x_3 + x_4 \leq 60 + M(1 - y_3),$$
$$y_1 + y_2 + y_3 \geq 2, \qquad y_1, y_2, y_3 \in \{0, 1\}.$$

  (4 Points)

Question 7: Mixed Binary Integer Programming – Product Mix (20 Points)

- Correctly defines continuous production variables $x_i \geq 0$ and binary setup variables $y_i \in \{0, 1\}$ for products $i = 1, \ldots, 4$. **(3 Points)**
- Correctly formulates the profit-maximization objective including marginal revenues and start-up costs:

$$\max\ 70x_1 + 60x_2 + 90x_3 + 80x_4 - 50000y_1 - 40000y_2 - 70000y_3 - 60000y_4.$$

  (3 Points)
- Correctly imposes the start-up linking constraints

$$0 \leq x_i \leq My_i, \qquad i = 1, \ldots, 4.$$

  (3 Points)
- Correctly models that no more than two products can be produced:

$$y_1 + y_2 + y_3 + y_4 \leq 2.$$

  (2 Points)
- Correctly models that Product 3 or Product 4 can be produced only if Product 1 or Product 2 is produced:

$$y_3 \leq y_1 + y_2, \qquad y_4 \leq y_1 + y_2.$$

  (3 Points)
- Correctly models that exactly one of the two resource constraints holds using a valid big-M formulation and an auxiliary binary variable. **(3 Points)**
- Provides a valid Excel Solver setup or screenshot and reports the optimal solution

$$x_2^* = 2000, \qquad y_2^* = 1,$$

  with all other production and setup variables equal to 0, and maximum total profit 80,000. **(3 Points)**

Question 8: Expected Value of a Discrete Random Variable (10 Points)

- Correctly computes
$$E[X] = 0(0.2) + 1(0.5) + 2(0.3) = 1.1.$$
(4 Points)

- Correctly computes
$$E[X^2] = 0^2(0.2) + 1^2(0.5) + 2^2(0.3) = 1.7.$$
(2 Points)

- Correctly uses linearity of expectation:
$$E[2X^2 + 3] = 2E[X^2] + 3 = 6.4.$$
(4 Points)